Alberto Luigi Fasulo

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Profile

MSc Electrical Engineering and Information Technology student at ETH Zurich specializing in control systems and machine learning, with hands-on experience in reinforcement learning, model predictive control, autonomous robotics, and embedded and wireless systems. Seeking an internship in robotics, control, ML, or autonomous systems from February 2027.

Education

ETH Zurich Zurich, Switzerland
MSc in Electrical Engineering and Information Technology; Control Systems and ML *2025 – Present*

- Relevant MSc coursework (completed and current): Computational Control, Dynamic Programming & Optimal Control, Recursive Estimation, Linear System Theory, Autonomous Mobile Robots, Probabilistic Artificial Intelligence, Introduction to Machine Learning, Image Analysis & Computer Vision, Game Theory & Control.

ETH Zurich Zurich, Switzerland
BSc in Electrical Engineering and Information Technology; Control Systems *2022 – 2025*

- Relevant coursework: Signal & System Theory I/II, Numerical Methods, Probability & Statistics, Python for Science & Machine Learning, Quad-Rotors: Control & Estimation.

Experience

Teaching Assistant, Computer Science I Feb 2025 – Aug 2026
ETH Zurich *Zurich, Switzerland*

- Delivered weekly tutorials for 20 students, supporting programming fundamentals, debugging, and problem-solving.
- Graded programming submissions for correctness and code clarity and supported exam administration.

Selected Projects

Deep Reinforcement Learning for Dynamic Flexible Job-Shop Scheduling | Python, PyTorch, PPO | GitHub Mar 2026 – Jul 2026

- ETH Zurich semester project (grade: 5.5/6). Developed a PPO-based scheduler for dynamic flexible job-shop scheduling with joint sequencing and routing decisions; benchmarked the learned policy against heuristic baselines.

Rocket Lander Model Predictive Control | MPC, PID, Constrained Optimization Jun 2026

- Redesigned a rocket lander controller from decoupled PID to constrained MPC with hard thrust and nozzle constraints, online equilibrium updates, and a 7 s prediction horizon.

Autonomous Quadrotor Control | MATLAB/Simulink, C++, PID, LQR Spring 2026

- Derived and simulated N-rotor dynamics and tuned PID and LQR controllers for hover and x-y position regulation in a Crazyflie-style quadrotor setup.

Wavvy – Personal Project | Django, KMM, BLE, Android, iOS, Redis Aug 2025 – Present

- Built a BLE proximity application using short-lived identifiers, Django backend services, Redis, shared Kotlin Multiplatform logic, and native Android/iOS clients.

Wi-Body – BSc Thesis | C, ESP32, Wi-Fi, Wireless Sensing | GitHub Jan 2025 – Jun 2025

- BSc thesis (grade: 5.5/6). Developed ESP32-based firmware and a wireless body-area sensing platform focused on synchronization, multi-node communication, low-power operation, and high throughput.

Raspberry Pi Robotic Arm | Python, Raspberry Pi, GPIO, Ultrasonic Sensing | GitHub 2022

- Built and programmed a five-servo robotic arm using Raspberry Pi and Python, integrating ultrasonic distance sensing for sensor-triggered automated movement and manual control.

Technical Skills

Programming: Python, C, C++, MATLAB/Simulink
ML/AI: PyTorch, Stable-Baselines3, PPO, reinforcement learning, computer vision
Control & Robotics: MPC, LQR, PID, optimal control, state estimation, autonomous systems, simulation
Embedded & Wireless: Raspberry Pi, ESP32, Arduino, GPIO, BLE, Wi-Fi, firmware development
Tools: Git, Linux, Docker, LaTeX
Languages: Italian (native), English (C1), German (B2), French (B1)